

Lumbar MRI Dataset for Disc Herniation Classification

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Overview

This dataset comprises lumbar spine MRI images used for binary classification of disc conditions into Normal and Herniated categories, supporting research in medical image analysis.

Data Collection

All data were anonymized and used for research purposes only. The dataset consists of a total of 1,426 MRI samples collected from multiple sources. A subset of 800 samples was obtained from publicly available medical imaging repositories, including platforms such as Kaggle, the Open Access Series of Imaging Studies (OASIS), and other open-access medical imaging sources. In addition, 258 samples were collected from a private hospital, while the remaining 368 samples were obtained from a governmental hospital. All data were aggregated to ensure diversity and variability in lumbar spine MRI cases, covering both normal and herniated conditions.

Dataset Composition

- Total: 1426
- Normal: 671
- Herniated: 755

Preprocessing

- Grayscale conversion to standardize image representation
- Resizing to a fixed resolution (200×200) for consistency
- Bilateral filtering to reduce noise while preserving structural details
- Pixel intensity normalization to the range $[0, 1]$ for stable model training

Usage

This dataset is used to evaluate the performance of classical machine learning models, including LightGBM, Random Forest, and SVM, with PCA for dimensionality reduction.

Purpose

The purpose of this dataset is to support research in medical image classification by enabling the development and evaluation of efficient classical machine learning approaches.

Key Notes

- Lightweight and computationally efficient pipeline
- PCA-based dimensionality reduction for handling high-dimensional data
- Focus on classical machine learning methods instead of deep learning
- Balanced trade-off between performance and computational cost
- Suitable for research, benchmarking, and educational purposes

License

This dataset is made available solely for research and educational purposes and should not be used for commercial applications.